exp 8.sci (C:\Users\pachi\Documents\exp 8.sci) - SciNotes

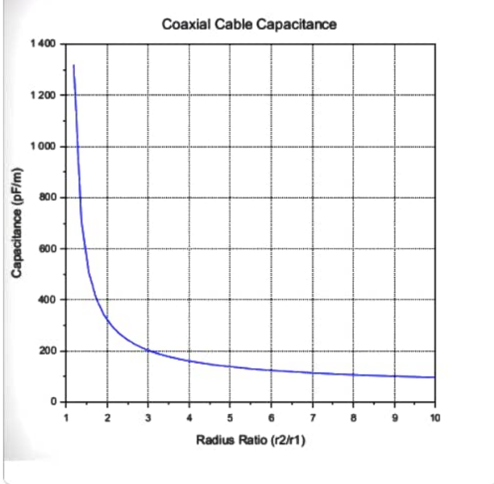
exp 6.sci | exp 7.sci | exp 8.sci

```
//------------------------
// Constants
//------------------------
pi = 3.1416;
e0 = 8.854e-12;

//------------------------
// Given Values
//------------------------
r1 = 3e-3;      // 3 mm = 0.003 m
r2 = 6e-3;      // 6 mm = 0.006 m
er = 4;

//------------------------
// Calculation
//------------------------
e = e0 * er;
ratio = r2 / r1;

c = (2 * pi * e) / log(ratio);

disp("Capacitance of Coaxial Cable");
disp(c,"Capacitance (F/m) =");
disp(c*1e12,"Capacitance (pF/m) =");

//------------------------
// Graph
//------------------------
x = linspace(1,10,50);      // Radius ratio
y = (2*pi*e)./log(x);

figure(1);
plot(x,y*1e12);
xlabel("Radius Ratio (r2/r1)");
ylabel("Capacitance (pF/m)");
title("Coaxial Cable Capacitance");
xgrid();
```

Scilab 2025.1.0 Console

```
 "Capacitance of Coaxial Cable"
  3.210D-10
 "Capacitance (F/m) ="
  321.03688
 "Capacitance (pF/m) ="

-->
```

Graphic window number 1



ex : 8

radius 1 = 3 mm

radius 2 = 6 mm

er = 4

Relative permitivity

$$\varepsilon = \varepsilon_0 \times e_r$$

$$= (8.854 \times 10^{-12}) \times 4$$

$$= 3.5416 \times 10^{-11} F/m$$

Radius ratio

$$r = \frac{r_2}{r_1} = \frac{0.006}{0.003} = 2$$

Neutral logarithm

$$\ln(2) = 0.6931$$

O/P

capacitance per unit

$$C = \frac{2\pi (3.5416 \times 10^{-11})}{0.6931}$$

$$= 3.21 \times 10^{-10} F/m$$

$$C = 321 pF/m$$